

Lesson 2-6: Divide Multi-Digit Numbers Using an Algorithm

Grade 6 Mathematics · Standard 6.NOS.2

Name: _____ Date: _____

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

Warm up with the lesson's core skills — one best answer each.

Question 1 SELECTED RESPONSE · 1 POINTWhat is $936 \div 12$?

- ☐ A 68
- ☐ B 76
- ☐ C 78
- ☐ D 87

Question 2 SELECTED RESPONSE · 1 POINTWhat is $2,485 \div 5$?

- ☐ A 487
- ☐ B 496
- ☐ C 497
- ☐ D 507

Question 3 SELECTED RESPONSE · 1 POINTWhat is $756 \div 9$?

- ☐ A 83
- ☐ B 84
- ☐ C 86
- ☐ D 94

Question 4 SELECTED RESPONSE · 1 POINT

What is $1,125 \div 9$?

- ☐ A 115
- ☐ B 124
- ☐ C 125
- ☐ D 135

Question 5 SELECTED RESPONSE · 1 POINT

A warehouse has 2,184 items to pack into boxes of 14. How many boxes are needed?

- ☐ A 146 boxes
- ☐ B 155 boxes
- ☐ C 156 boxes
- ☐ D 166 boxes

Question 6 SELECTED RESPONSE · 1 POINT

What is $4,896 \div 12$?

- ☐ A 48
- ☐ B 380
- ☐ C 408
- ☐ D 480

Part B — Test-Format Questions

These match the state test's formats: two-part questions, select-all, and written responses.

Question 7**TWO-PART QUESTION****Part A**

Use the long division algorithm to find $936 \div 4$.

Fill in the bubble next to the one best answer.

- ☐ (A) 209
- ☐ (B) 232
- ☐ (C) 234
- ☐ (D) 243

Part B

Answer Part A before Part B — Part B asks about your reasoning.

In the D-M-S-B cycle, what does the SUBTRACT step tell you?

- ☐ (A) How many times the divisor fits into the whole dividend.
- ☐ (B) How much of the current value is left over before the next digit is brought down.
- ☐ (C) Whether the quotient will be even.
- ☐ (D) The final remainder of the entire problem.

Question 8**WRITTEN RESPONSE · 2 POINTS****Type II Reasoning: The Dropped Zero**

Andre computed $618 \div 3$. He wrote: ' $6 \div 3 = 2$, then $1 \div 3$ does not go so I skip it, then $8 \div 3 = 2$ remainder 2. My answer is 22 remainder 2.'

Explain Andre's misconception and give the correct quotient.

Write your answer in complete sentences. Show or explain your mathematical thinking.

Part C — Show Your Thinking

Answer in writing, the way the state test's constructed responses work.

Question 9 WRITTEN RESPONSE · 2 POINTS

The station must load 3,360 ration packs equally into 14 storage pods. Solve it with long division. Name each step — divide, multiply, subtract, bring down — and check your answer by multiplying.

Write your answer in complete sentences. Show or explain your mathematical thinking.

When you finish, check your reasoning with your teacher or family. Your teacher has the scoring guide for the written responses.